

Height-Resolved Curvature Decomposition: Separating Real Stability Curvature from Coordinate Effects

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The 2D contour diagnostic panel demonstrates that sharp vertical profile features near the critical Richardson number ($Ri_c \approx 0.20$) often stem from coordinate stretching rather than purely local thermodynamic phase changes. When turbulent flux varies with height, the local Obukhov length scale $L(z)$ distorts the vertical coordinate frame.

We decompose the total profile curvature of $Ri(z)$ into distinct physical and numerical components:

$$Ri_{zz} = \underbrace{Ri_{\zeta\zeta}\zeta_z^2}_{C_{\text{const}}} + \underbrace{Ri_{\zeta}\zeta_{zz}}_{C_{\text{coord}}} + E_{\text{error}}$$

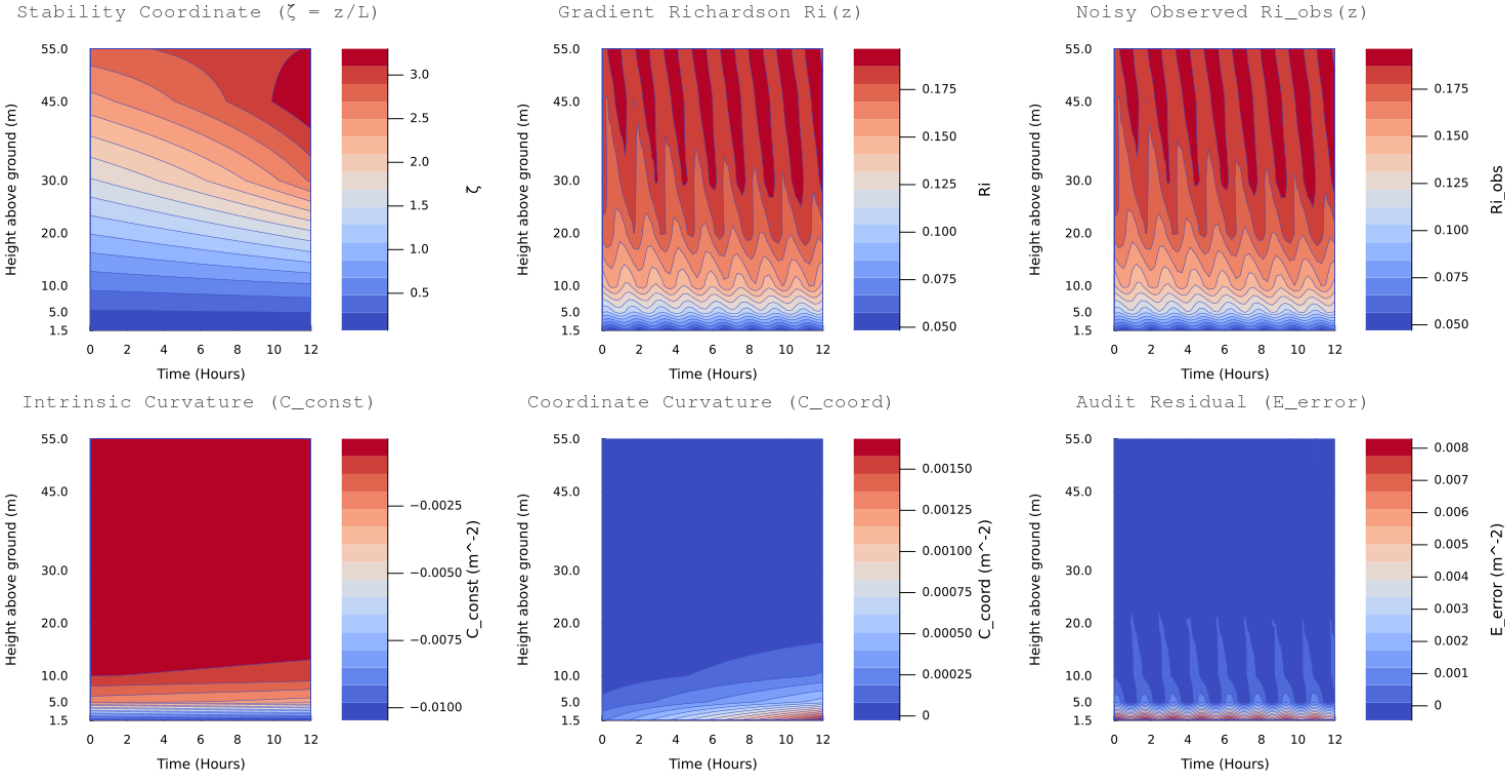
- **Intrinsic Stability Curvature (C_{const}):** Expected curvature from local thermodynamic stability holding the similarity coordinate fixed. Under Monin–Obukhov similarity ($\beta_m = 5.0, \beta_h = 5.0$), this term remains negative, reflecting gradual lower-level stabilization.
- **Coordinate-Stretching Curvature (C_{coord}):** Curvature introduced because $\zeta(z) = z/L(z)$ varies with height under flux-divergent SBL conditions. Near nocturnal low-level jet (LLJ) noses, this coordinate effect spikes strongly positive, counteracting intrinsic cooling.
- **Audit Residual (E_{error}):** Quantifies operator truncation error ($\mathbf{D}_2 Ri_{\text{smooth}} - Ri_{\text{exact}}$). Keeping numerical errors separate prevents finite-difference artifacts on non-uniform CASES-99 tower grids (1.5 m to 55 m) from contaminating physical curvature fields.

Key Physical Insights (12-Hour SBL Cooling Cycle)

- **Near-Surface Cancellation ($z \approx 10$ m):** During deep stable hours (Hours 6–12), positive coordinate curvature ($C_{\text{coord}} \approx +0.0036 \text{ m}^{-2}$) cancels negative stability curvature ($C_{\text{const}} \approx -0.0040 \text{ m}^{-2}$). The resulting locally linear profile ($Ri_{zz} \approx -0.0004 \text{ m}^{-2}$) hides active, offsetting physical processes.

- **Noise Suppression & Filtering:** Synthetic sensor jitter ($\sigma_{\text{obs}} = 0.008$) added to Ri_{obs} is smoothed via Tikhonov-regularized differentiation ($\lambda_{\text{reg}} = 5.0$) before operator evaluation, isolating physical signal from observational noise.

GSPT 2D Curvature & Profile Audit: 12-Hour SBL Cooling Cycle (CASES-99, $\beta_m = 5.0$, $\beta_h = 5.0$, $Ri_c = 0.20$)



6-Panel Diagnostic Layout

Panel Position	Variable	Physical / Numerical Interpretation
Top Left	Stability Coordinate (ζ)	Dimensionless height ratio $z/L(z)$ across time and altitude.
Top Center	Gradient Richardson (Ri)	Exact analytical profile derived from SBL similarity theory.
Top Right	Observed Richardson (Ri_{obs})	Synthetic sensor data containing Gaussian observational jitter.
Bottom	Intrinsic Curvature (	Pure thermodynamic curvature assuming constant

Left	C_{const}	coordinate scaling.
Bottom Center	Coordinate Curvature (C_{coord})	Curvature generated by vertical flux divergence and LLJ shear.
Bottom Right	Audit Residual (E_{error})	Explicit finite-difference discretization residual ($\mathbf{D}_2 Ri_{\text{smooth}} - Ri_{\text{exact}}$).

Code:

```
#!/usr/bin/env julia
# src/gspt_curvature_contour_v3.jl
# version 3 of the GSPT curvature contour plotting script
# David E. England, Ph.D.
# email: david.england@uah.edu
using LinearAlgebra
using Printf

# Try to use Plots.jl
try
    using Plots
catch e
    @warn "Plots.jl package not found in current environment. Script will save numerical data instead of plotting."
end

function stencil_weights(z_stencil::Vector{Float64}, z0::Float64, m::Int)
    p = length(z_stencil)
    A = [Float64(z - z0)^(k - 1) / factorial(k - 1) for k in 1:p, z in z_stencil]
    b = zeros(p)
    b[m + 1] = 1.0 # Select target derivative order
    return A \ b
end

function build_operators(z::Vector{Float64})
    n = length(z)
    D1 = zeros(n, n)
    D2 = zeros(n, n)
```

```

for i in 1:n
    if i == 1
        idx = [1, 2, 3]
    elseif i == n
        idx = [n-2, n-1, n]
    else
        idx = [i-1, i, i+1]
    end

    D1[i, idx] = stencil_weights(z[idx], z[i], 1)
    D2[i, idx] = stencil_weights(z[idx], z[i], 2)
end

return D1, D2
end

Ri_model( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ ) =  $\zeta * (1.0 + \beta_h * \zeta) / (1.0 + \beta_m * \zeta)^2$ 

function Ri_zeta( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ )
    return  $(1.0 + (2.0 * \beta_h - \beta_m) * \zeta) / (1.0 + \beta_m * \zeta)^3$ 
end

function Ri_zetazeta( $\zeta$ ,  $\beta_m$ ,  $\beta_h$ )
    num =  $2.0 * \beta_h - 4.0 * \beta_m + 2.0 * \beta_m * (\beta_m - 2.0 * \beta_h) * \zeta$ 
    return num /  $(1.0 + \beta_m * \zeta)^4$ 
end

function plot_gspt_curvature_layers(
    hours::Vector{Float64},
    z_tower::Vector{Float64},
    zeta_mat::Matrix{Float64},
    ri_smooth_mat::Matrix{Float64},
    ri_obs_mat::Matrix{Float64},
    C_const::Matrix{Float64},
    C_coord::Matrix{Float64},
    E_error::Matrix{Float64};
    save_path::String = "gspt_curvature_contour_v3.png"
)
    if !@isdefined Plots
        @warn "Plots.jl not loaded. Skipping PNG rendering."
    end
end

```

```

    return nothing
end

gr() # Set GR backend

# Common plotting parameters
plt_opts = (
    xlabel = "Time (Hours)",
    ylabel = "Height above ground (m)",
    yticks = (z_tower, [@sprintf("%.1f", h) for h in z_tower]),
    fill = true,
    c = :coolwarm,
    linewidth = 0.5,
    tickfont = font(8, "sans-serif"),
    guidefont = font(9, "sans-serif"),
    titlefont = font(10, "sans-serif bold")
)

# Row 1 plots
p1 = contourf(hours, z_tower, zeta_mat;
    title="Stability Coordinate ( $\zeta = z/L$ )",
    colorbar_title=" $\zeta$ ",
    plt_opts...)

p2 = contourf(hours, z_tower, ri_smooth_mat;
    title="Gradient Richardson  $Ri(z)$ ",
    colorbar_title="Ri",
    plt_opts...)

p3 = contourf(hours, z_tower, ri_obs_mat;
    title="Noisy Observed  $Ri_{obs}(z)$ ",
    colorbar_title="Ri_obs",
    plt_opts...)

# Row 2 plots
p4 = contourf(hours, z_tower, C_const;
    title="Intrinsic Curvature ( $C_{const}$ )",
    colorbar_title="C_const ( $m^{-2}$ )",
    plt_opts...)

p5 = contourf(hours, z_tower, C_coord;

```

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        title="Coordinate Curvature (C_coord)",
        colorbar_title="C_coord (m-2)",
        plt_opts...)

p6 = contourf(hours, z_tower, E_error;
              title="Audit Residual (E_error)",
              colorbar_title="E_error (m-2)",
              plt_opts...)

# Assemble into a 2x3 diagnostic panel layout
full_plot = plot(p1, p2, p3, p4, p5, p6,
                 layout = (2, 3),
                 size = (1500, 800),
                 plot_title = "GSPT 2D Curvature & Profile Audit: 12-Hour SBL Cool:
                 plot_titlefont = font(12, "sans-serif bold"),
                 margin = 6Plots.mm)

savefig(full_plot, save_path)
println("Successfully rendered and saved: $save_path")
return full_plot
end

function run_test_simulation()
    println("Setting up SCM 12-hour cooling cycle...")

    # CASES-99 55m Tower heights
    z_tower = [1.5, 5.0, 10.0, 20.0, 30.0, 45.0, 55.0]
    n_z = length(z_tower)
    D1, D2 = build_operators(z_tower)

    # Time steps: 50 points over 12 hours
    hours = collect(range(0.0, 12.0, length=50))
    n_t = length(hours)

    # Curvature & profile matrices
    zeta_mat = zeros(n_z, n_t)
    ri_smooth_mat = zeros(n_z, n_t)
    ri_obs_mat = zeros(n_z, n_t)
    C_const_mat = zeros(n_z, n_t)
    C_coord_mat = zeros(n_z, n_t)
    E_error_mat = zeros(n_z, n_t)

```

```

# SBL model settings (using user-requested grassland parameters)
 $\beta_m$ ,  $\beta_h$  = 5.0, 5.0
L0 = 20.0
sigma_obs = 0.008
lambda_reg = 5.0

for t_idx in 1:n_t
    t = hours[t_idx]

    # Non-stationary LLJ Jet nose intensity growing with cooling
    amp = 0.1 + 0.35 * (t / 12.0)
    L_profile = [L0 * (1.0 - amp * sin( $\pi$  * zi / 60.0)) for zi in z_tower]

    # Similarity coordinate profile
     $\zeta$  = z_tower ./ L_profile
     $\zeta_z$  = D1 *  $\zeta$ 
     $\zeta_{zz}$  = D2 *  $\zeta$ 

    zeta_mat[:, t_idx] =  $\zeta$ 

    # Analytical GSPT curvature
    ri_z = [Ri_zeta( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z]
    ri_zz = [Ri_zetazeta( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z]

    C_const = ri_zz .* ( $\zeta_z$  .^ 2)
    C_coord = ri_z .*  $\zeta_{zz}$ 
    Ri_exact = C_const .+ C_coord

    C_const_mat[:, t_idx] = C_const
    C_coord_mat[:, t_idx] = C_coord

    # Add random sensor jitter (using deterministic seed offsets for repeatability)
    noise = sigma_obs .* sin.(t_idx .+ (1:n_z) .* 1.5)
    Ri_obs = [Ri_model( $\zeta$ [i],  $\beta_m$ ,  $\beta_h$ ) for i in 1:n_z] .+ noise
    ri_obs_mat[:, t_idx] = Ri_obs

    # Track A: Tikhonov regularization filter
    R = D2' * D2
    A_reg = I(n_z) + lambda_reg .* R
    Ri_smooth = A_reg \ Ri_obs

```

```

    ri_smooth_mat[:, t_idx] = Ri_smooth

    # Computed physical curvature
    M_Ri_zz = D2 * Ri_smooth

    # Discrete audit error
    E_error_mat[:, t_idx] = M_Ri_zz .- Ri_exact
end

println("Matrices populated successfully. Generating Plots panel...")

plot_gspt_curvature_layers(hours, z_tower, zeta_mat, ri_smooth_mat, ri_obs_mat,
                           C_const_mat, C_coord_mat, E_error_mat;
                           save_path = "gspt_curvature_contour_v3.png")

end

run_test_simulation()

```

The script sets up a complete synthetic boundary layer experiment designed to evaluate how numerical operators handle non-uniform tower grids, flux-divergent jet dynamics, observational jitter, and spatial filtering.

1. Non-Uniform Stencil Differentiation ($\mathbf{D}_1, \mathbf{D}_2$)

- **Method:** Computes derivative weights by solving a local Taylor expansion system $\mathbf{A}\mathbf{w} = \mathbf{b}$ for derivative orders $m = 1$ ($\mathbf{D}_1$) and $m = 2$ ($\mathbf{D}_2$).
- **Grid Adaptation:** Because CASES-99 tower levels are irregularly spaced, ranging from 1.5 m near the surface to 10 m spacing aloft, standard uniform finite differences introduce severe errors. The code applies 3-point centered stencils on interior nodes and one-sided 3-point stencils at the top and bottom boundaries.

2. Synthetic Low-Level Jet (LLJ) Signature

- **Mechanism:** Prescribes a synthetic Obukhov length field $L(z, t)$ whose vertical structure mimics the stability signature associated with a developing nocturnal LLJ, modulated over the 12-hour cooling period:

$$L(z, t) = L_0 \left[1 - \left(0.1 + 0.35 \frac{t}{12} \right) \sin \left(\frac{\pi z}{60} \right) \right]$$

- **Physical Effect:** Produces a $Ri(z)$ profile with the curvature signature typical of a developing nocturnal LLJ nose around $z \approx 30\text{--}45$ m. This introduces vertical flux divergence ($\zeta_{zz} \neq 0$), which generates the positive coordinate curvature (C_{coord}) that competes with surface cooling (C_{const}).

3. Observational Noise Jitter

- **Formula:** $Ri_{\text{obs}}(z, t) = Ri_{\text{exact}}(z, t) + \sigma_{\text{obs}} \sin(t_{\text{idx}} + 1.5z_{\text{idx}})$
- **Purpose:** Injects continuous sensor noise ($\sigma_{\text{obs}} = 0.008$) into exact similarity fields. Using deterministic sine offsets mimics high-frequency instrument jitter while keeping the benchmark strictly reproducible across runs.

4. Tikhonov Regularization (Smoothing)

- **Operator:** Solves the linear system $(\mathbf{I} + \lambda_{\text{reg}} \mathbf{D}_2^T \mathbf{D}_2) Ri_{\text{smooth}} = Ri_{\text{obs}}$ using regularization parameter $\lambda_{\text{reg}} = 5.0$.
- **Function:** Taking raw numerical derivatives of noisy data catastrophically amplifies high-frequency jitter. The curvature penalty term ($\mathbf{D}_2^T \mathbf{D}_2$) smooths Ri_{obs} prior to second differentiation, allowing the audit residual E_{error} to isolate discretization errors rather than sensor noise.

Quick Parameter Tuning Guide

- **Change Tower Levels (`z_tower`):** Modify the values inside `[1.5, 5.0, ...]` to match actual sensor heights (e.g., add `70.0` or `100.0` for taller boundary layer masts).
- **Adjust SBL Regimes (`beta_m` , `beta_h`):** Set `beta_m = 4.7` and `beta_h = 4.7` for standard Businger–Dyer formulations, lower to `4.0` for weakly stable conditions agreeing with a classical critical value of `0.25` , or increase to `5.0` or more for strongly stable nocturnal conditions.
- **Control Noise Jitter (`sigma_obs`):** Increase `sigma_obs` to `0.02` to test high-noise sonic anemometer environments.
- **Filter Jitter Amplification (`lambda_reg`):** If derivative fields look noisy, increase `lambda_reg` (e.g., `10.0 – 20.0`). If the filter oversmooths physical gradients, reduce it toward `1.0` .